Electrical & Computer Engineering
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Yuyang Wang

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Appointment

University of Connecticut Storrs, CT, USA
Assistant Professor, Department of Electrical and Computer Engineering 2025–Present

Columbia University in the City of New York New York, NY, USA
Postdoctoral Research Scientist, Columbia Nano Initiative 2021–2025

Education

University of California, Santa Barbara Santa Barbara, CA, USA
Ph.D. in Electrical and Computer Engineering 2018–2021

University of California, Santa Barbara Santa Barbara, CA, USA
M.S. in Electrical and Computer Engineering 2015–2018

Tsinghua University Beijing, China
B.Eng. in Electronic Engineering 2011–2015

Publication

Book Chapters

B1 **Y. Wang**, X. Meng, M. Lipson, A. L. Gaeta, and K. Bergman, "Co-packaged optics (CPO) for communications," in *Optical Fiber Telecommunications VIII*, A. E. Willner, Ed., in press. Academic Press, 2026

Journal Articles

J1 Z. Wu, **Y. Wang**, and K. Bergman, "SiPAM: Silicon Photonic Accelerated Memory Pooling Architecture," *IEEE Micro*, pp. 1–10, 2026. 10.1109/MM.2026.3670009

J2 R. Parsons, K. Jang, **Y. Wang**, A. Novick, A. M. Smith, C. C. Tison, Y. Gebregiorgis, V. Deenadayalan, M. van Niekerk, L. Carpenter, T. Ngai, G. Leake, D. Coleman, X. Meng, S. Preble, M. L. Fanto, K. Bergman, and A. Rizzo, "Highly uniform thermally undercut silicon photonic devices in a 300 mm CMOS foundry process," *Scientific Reports*, vol. 15, no. 1, p. 29 906, Aug. 2025. 10.1038/s41598-025-14480-4

J3 R. Parsons, A. Oh, J. Robinson, S. Wang, M. Cullen, K. Jang, A. James, **Y. Wang**, and K. Bergman, "Foundry-enabled wafer-scale characterization and modeling of silicon photonic DWDM links," *Nanophotonics*, vol. 14, no. 27, pp. 5363–5374, 2025. 10.1515/nanoph-2025-0439

J4 J. Robinson, R. Parsons, **Y. Wang**, K. Jang, X. Meng, and K. Bergman, "Silicon photonic DWDM micro-resonator link initialization under fabrication variation," *Optics Express*, vol. 33, no. 26, pp. 53 700–53 713, Dec. 2025. 10.1364/OE.577064

J5 S. Wang, **Y. Wang**, S. Sanyal, R. Parsons, X. Ji, Y. Okawachi, X. Meng, M. Lipson, A. Gaeta, and K. Bergman, "Automated tuning of a ring-assisted MZI-based interleaver for Kerr frequency combs," *Optics Letters*, vol. 50, no. 2, p. 698, Jan. 2025. 10.1364/OL.546772

J6 **Y. Wang**, S. Wang, R. Parsons, S. Sanyal, V. Gopal, A. Novick, A. Rizzo, M. Lipson, A. L. Gaeta, and K. Bergman, "Co-Designed Silicon Photonics Chip I/O for Energy-Efficient Petascale Connectivity," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 15, no. 8, pp. 1581–1591, Aug. 2025. 10.1109/TCPMT.2024.3492189 Best Paper of the Year

J7 Z. Wu, L. Yuan Dai, **Y. Wang**, S. Wang, and K. Bergman, "Flexible silicon photonic architecture for accelerating distributed deep learning," *Journal of Optical Communications and Networking*, vol. 16, no. 2, A157, Feb. 2024. 10.1364/JOCN.497372

J8 A. James, A. Rizzo, **Y. Wang**, A. Novick, S. Wang, R. Parsons, K. Jang, M. Hattink, and K. Bergman, "Process Variation-Aware Compact Model of Strip Waveguides for Photonic Circuit Simulation," *Journal of Lightwave Technology*, pp. 1–14, 2023. 10.1109/JLT.2023.3238847

J9 A. Novick, A. James, L. Y. Dai, Z. Wu, A. Rizzo, S. Wang, **Y. Wang**, M. Hattink, V. Gopal, K. Jang, R. Parsons, and K. Bergman, "High-bandwidth density silicon photonic resonators for energy-efficient optical interconnects," *Applied Physics Reviews*, vol. 10, no. 4, p. 041 306, Dec. 2023. 10.1063/5.0160441

J10 **Y. Wang**, P. Sun, J. Hulme, M. A. Seyedi, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Energy Efficiency and Yield Optimization for Optical Interconnects via Transceiver Grouping," *Journal of Lightwave Technology*, vol. 39, no. 6, pp. 1567–1578, Mar. 2021. 10.1109/JLT.2020.3039489

J11 Z. Zhang, R. Wu, **Y. Wang**, C. Zhang, E. J. Stanton, C. L. Schow, K.-T. Cheng, and J. E. Bowers, "Compact Modeling for Silicon Photonic Heterogeneously Integrated Circuits," *Journal of Lightwave Technology*, vol. 35, no. 14, pp. 2973–2980, Jul. 2017. doi 10.1109/JLT.2017.2706721

Conference Proceedings

C1 M. Cullen, S. Sanyal, S. Wang, K. J. McNulty, **Y. Wang**, M. Lipson, A. L. Gaeta, and K. Bergman, "Reconfigurable Kerr-comb for enhanced link margins in >1 Tb/s direct-drive DWDM links," in *Conference on Lasers and Electro-Optics (CLEO)*, Charlotte, NC: Optica Publishing Group, 2026, STh4A.1 Highlighted

C2 B. C. George, W. Wang, Z. Wu, **Y. Wang**, X. Meng, M. Ghobadi, and K. Bergman, "Reconfiguration-aware direct-connect AI cluster using spatial-and-wavelength selective switching," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2026, M3F.4. doi 10.1364/OFC.2026.M3F.4 Top-Scored

C3 S. Sanyal, M. Cullen, S. Wang, **Y. Wang**, Y. Okawachi, K. J. McNulty, M. Lipson, K. Bergman, and A. L. Gaeta, "High-power Kerr comb source for data communications," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2026, Th1F.5. doi 10.1364/OFC.2026.Th1F.5

C4 S. Wang, M. Cullen, T. Zypman, S. Sanyal, K. J. McNulty, **Y. Wang**, X. Meng, M. Lipson, A. L. Gaeta, and K. Bergman, "On-chip programmable MZI-based Fourier synthesizer for ultra-broadband Kerr-comb equalization," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2026, M1E.4. doi 10.1364/OFC.2026.M1E.4

C5 N. Nauman, J. Robinson, **Y. Wang**, K. Jang, X. Meng, and K. Bergman, "Photonic Analog-to-Digital Architecture for Accelerating Multiply-Accumulate Operations," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2025, W3D.5. doi 10.1364/OFC.2025.W3D.5

C6 R. Parsons, S. Sanyal, M. Cullen, **Y. Wang**, A. Novick, X. Ji, Y. Okawachi, X. Meng, M. Lipson, A. Gaeta, and K. Bergman, "Dispersion-Engineered Resonator-Based Interleaver Co-Designed with Kerr Comb Source," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2025, Tu2G.6. doi 10.1364/OFC.2025.Tu2G.6

C7 A. Rovinski, Y. Ou, C. Ou, D. Khilwani, **Y. Wang**, S. Wang, S. Lee, K. Bergman, A. Molnar, and C. Batten, "Scaling Co-Packaged Optical Interconnects Using Hybrid 2.5D/3D Integration," in *IEEE International Symposium on Circuits and Systems (ISCAS)*, May 2025, pp. 1–5. doi 10.1109/ISCAS56072.2025.11043946

C8 **Y. Wang**, S. Wang, S. Sanyal, N. Nauman, R. Parsons, J. Robinson, M. Hattink, K. Jang, A. Novick, K. J. McNulty, X. Meng, M. Lipson, A. L. Gaeta, and K. Bergman, "Order-Preserving Channel Calibration of Kerr Comb–Driven Microresonator-Based DWDM Link," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2025, W1E.1. doi 10.1364/OFC.2025.W1E.1

C9 A. Novick, M. Hattink, A. Rizzo, **Y. Wang**, V. Gopal, S. Wang, R. Parsons, and K. Bergman, "Integrated Photonic Resonant Modulator-Based Equalization and Optimization for DWDM," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2024, W1K.5. doi 10.1364/OFC.2024.W1K.5

C10 S. Wang, **Y. Wang**, X. Meng, K. Hosseini, T. T. Hoang, and K. Bergman, "Automated Tuning of Ring-Assisted MZI-Based Interleaver for DWDM Systems," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2024, Th1A.3. doi 10.1364/OFC.2024.Th1A.3

C11 **Y. Wang**, S. Wang, R. Parsons, A. Novick, V. Gopal, K. Jang, A. Rizzo, C.-P. Chiu, K. Hosseini, T. T. Hoang, S. Shumarayev, and K. Bergman, "Silicon Photonics Chip I/O for Ultra High-Bandwidth and Energy-Efficient Die-to-Die Connectivity," in *IEEE Custom Integrated Circuits Conference (CICC)*, Denver, CO, USA: IEEE, Apr. 2024, pp. 1–8. doi 10.1109/CICC60959.2024.10529018

C12 Z. Wu, R. Parsons, S. Wang, **Y. Wang**, and K. Bergman, "Wavelength Reconfigurable Transceiver for Multi-Interface Compute Accelerator Networks," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2024, W4F.2. doi 10.1364/OFC.2024.W4F.2

C13 G. Michelogiannakis, Y. Arafa, B. Cook, L. Y. Dai, A.-H. Hameed Badawy, M. Glick, **Y. Wang**, K. Bergman, and J. Shalf, "Efficient Intra-Rack Resource Disaggregation for HPC Using Co-Packaged DWDM Photonics," in *IEEE International Conference on Cluster Computing (CLUSTER)*, Santa Fe, NM, USA: IEEE, Oct. 2023, pp. 158–172. doi 10.1109/CLUSTER52292.2023.00021

C14 S. Wang, A. Novick, A. Rizzo, R. Parsons, S. Sanyal, K. J. McNulty, B. Y. Kim, Y. Okawachi, **Y. Wang**, A. Gaeta, M. Lipson, and K. Bergman, "Integrated, Compact, and Tunable Band-Interleaving of a Kerr Comb Source," in *Conference on Lasers and Electro-Optics (CLEO)*, San Jose, CA: Optica Publishing Group, 2023, STh3J.6. doi 10.1364/CLEO_SI.2023.STh3J.6

C15 **Y. Wang**, A. Novick, R. Parsons, S. Wang, K. Jang, A. James, M. Hattink, V. Gopal, A. Rizzo, C.-P. Chiu, K. Hosseini, T. T. Hoang, and K. Bergman, "Scalable architecture for sub-pJ/b multi-Tbps comb-driven DWDM silicon photonic transceiver," in *Next-Generation Optical Communication: Components, Sub-Systems, and Systems XII*, G. Li, K. Nakajima, and A. K. Srivastava, Eds., San Francisco, United States: SPIE, Mar. 2023, p. 55. doi 10.1117/12.2649506

C16 **Y. Wang**, S. Wang, A. Novick, A. James, R. Parsons, A. Rizzo, and K. Bergman, "Dispersion-Engineered and Fabrication-Robust SOI Waveguides for Ultra-Broadband DWDM," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2023, Th3A.4. doi 10.1364/OFC.2023.Th3A.4

C17 A. James, **Y. Wang**, A. Rizzo, and K. Bergman, "Flexible, Process-Aware Compact Model of Effective Index in Silicon Waveguides for Commercial Foundries," in *International Conference on Numerical Simulation of Optoelectronic Devices (NUSOD)*, Turin, Italy: IEEE, Sep. 2022, pp. 173–174. doi 10.1109/NUSOD54938.2022.9894784

C18 **Y. Wang** and K.-T. Cheng, "Traffic-Adaptive Power Reconfiguration for Energy-Efficient and Energy-Proportional Optical Interconnects," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, Munich, Germany: IEEE, Nov. 2021, pp. 1–9. doi 10.1109/ICCAD51958.2021.9643475

C19 **Y. Wang**, J. Hulme, P. Sun, M. Jain, M. A. Seyedi, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Characterization and Applications of Spatial Variation Models for Silicon Microring-Based Optical Transceivers," in *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, USA: IEEE, Jul. 2020, pp. 1–6. doi 10.1109/DAC18072.2020.9218608

C20 **Y. Wang** and K.-T. Cheng, "Task Mapping-Assisted Laser Power Scaling for Optical Network-on-Chips," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, Westminster, CO, USA: IEEE, Nov. 2019, pp. 1–6. doi 10.1109/ICCAD45719.2019.8942146

C21 **Y. Wang**, M. A. Seyedi, J. Hulme, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Bidirectional tuning of microring-based silicon photonic transceivers for optimal energy efficiency," in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Tokyo, Japan: ACM, Jan. 2019, pp. 370–375. doi 10.1145/3287624.3287649

C22 **Y. Wang**, L. Shao, M. A. Lastras-Montano, and K.-T. Cheng, "Taming Emerging Devices' Variation and Reliability Challenges with Architectural and System Solutions [Invited]," in *IEEE International Conference on Microelectronic Test Structures (ICMTS)*, Kita-Kyushu City, Fukuoka, Japan: IEEE, Mar. 2019, pp. 90–95. doi 10.1109/ICMTS.2019.8730924

C23 **Y. Wang**, M. A. Seyedi, R. Wu, J. Hulme, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Energy-efficient channel alignment of DWDM silicon photonic transceivers," in *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Dresden, Germany: IEEE, Mar. 2018, pp. 601–604. doi 10.23919/DATE.2018.8342079

C24 R. Wu, M. A. Seyedi, **Y. Wang**, J. Hulme, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Pairing of microring-based silicon photonic transceivers for tuning power optimization," in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Jeju, South Korea: IEEE, Jan. 2018, pp. 135–140. doi 10.1109/ASPDAC.2018.8297295

C25 R. Wu, **Y. Wang**, Z. Zhang, C. Zhang, C. L. Schow, J. E. Bowers, and K.-T. Cheng, "Compact modeling and circuit-level simulation of silicon nanophotonic interconnects," in *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Lausanne, Switzerland: IEEE, Mar. 2017, pp. 602–605. doi 10.23919/DATE.2017.7927057

C26 A. Ghofrani, M. A. Lastras-Montaño, **Y. Wang**, and K.-T. Cheng, "In-place Repair for Resistive Memories Utilizing Complementary Resistive Switches," in *International Symposium on Low Power Electronics and Design (ISLPED)*, San Francisco Airport, CA, USA: ACM, Aug. 2016, pp. 350–355. doi 10.1145/2934583.2934590

C27 C. Xu, F. X. Lin, **Y. Wang**, and L. Zhong, "Automated OS-level Device Runtime Power Management," in *International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Istanbul, Turkey: ACM, Mar. 2015, pp. 239–252. doi 10.1145/2694344.2694360

Mentoring

U : undergraduate; **M** : master's; **D** : doctoral

Waseem Faheem **D** Ph.D. Student, University of Connecticut — Spring 2026 – Present

Zahin Zeal **D** Ph.D. Student, University of Connecticut — Fall 2026 – Present

Harvansh Pelia **U** Undergraduate Student, University of Connecticut — Spring 2026 – Present
Honors Thesis Supervision

Willow Pichardo **U** Undergraduate Student, University of Connecticut — Spring 2026 – Present

Uyen Bao Pham **U** Undergraduate Student, University of Connecticut — Spring 2026 – Present

Khang "James" Le **U** Undergraduate Student, University of Connecticut — Spring 2026 – Present

Abidur Rahman **U** Undergraduate Student, Wayne State University — Summer 2024
Summer Undergraduate Research Experience (SURE) Program of Columbia Engineering

Teaching

U : undergraduate; G : graduate

University of Connecticut Storrs, CT, USA

- Instructor — ECE 2001: Electrical Circuits U — Spring 2026
- Project Advisor — ECE 4901: Senior Design U — Spring 2026

Columbia University in the City of New York New York, NY, USA

- Guest Lecturer — ELEN 9404: Seminar in Lightwave Communications G — Spring 2024
- Guest Lecturer — ELEN 9403: Seminar in Photonics G — Spring 2023

University of California, Santa Barbara Santa Barbara, CA, USA

- Teaching Assistant — ECE 153B: Sensor & Peripheral Interface Design U — Winter 2019

Service

Guest Editor IEEE Solid-State Circuits Magazine 2027
Spring 2027 Issue: Silicon Photonics

Technical Program Committee Optical Fiber Communication Conference (OFC) 2027
N2: Optics and Photonics for Data Center and Computing Applications

Topical Chair IEEE Photonics Society Summer Topicals Meeting Series (SUM) 2026
Electronic-Photonic Systems at Scale — Architecture, Methodology, and Going 3D (EP3D)

Technical Program Committee IEEE Hot Interconnects Symposium (HotI) 2026

Technical Program Committee IEEE Chiplet Workshop: Standards, Circuits and Systems 2025
Chiplet Interconnects and Packaging

Ad-hoc Reviewer

- *Nature Photonics*, Springer Nature
- *Nature Nanotechnology*, Springer Nature
- *Nature Communications*, Springer Nature
- *npj Nanophotonics*, Springer Nature
- *Communications Engineering*, Springer Nature
- *APL Photonics*, American Institute of Physics
- *Photonics*, MDPI
- *ACM Transactions on Design Automation of Electronic Systems*, ACM
- *Journal of Lightwave Technology*, IEEE/Optica
- *Journal of Selected Areas in Communications*, IEEE
- *Journal on Emerging and Selected Topics in Circuits and Systems*, IEEE
- *IEEE Transactions on Computers*, IEEE
- *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, IEEE
- *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, IEEE
- *IEEE Access*, IEEE

Award and Honor

Best Paper Award IEEE Transactions on Components, Packaging and Manufacturing Technology 2025
"Co-Designed Silicon Photonics Chip I/O for Energy-Efficient Petascale Connectivity"

Presentation

Seminar Department of Physics, University of Connecticut — Storrs, CT, USA
Integrated Photonics Systems in the Package: Beyond Massive Parallel Data Movement — Mar. 2026

Seminar Department of ECE, University of Connecticut — Storrs, CT, USA
Photonics-Empowered Computing: From Efficient to Functional Data Movement — Feb. 2025

Seminar School of ECEE, Arizona State University — Tempe, AZ, USA
Photonics-Empowered Computing: From Efficient to Functional Data Movement — Feb. 2025

Seminar Department of ECE, University of Minnesota — Minneapolis, MN, USA
Photonics-Empowered Computing: From Efficient to Functional Data Movement — Feb. 2025

Presentation SRC TECHCON — Austin, TX, USA
Scalable Kerr-Comb Driven DWDM Silicon Photonics Chip I/O — Sep. 2024

Seminar School of EECS, Pennsylvania State University — University Park, PA, USA
Co-Designing Photonics for Heterogeneously Integrated Systems — Apr. 2024

Invited Talk SPIE Photonics West — San Francisco, CA, USA
Scalable Architecture for Sub-pJ/b Multi-Tbps Comb-Driven DWDM Silicon Photonic Transceiver — Jan. 2023

Poster Ph.D. Forum, ACM/IEEE Design Automation Conference (DAC) — Online Virtual Event
Design and Optimization of Variation-Aware Runtime-Reconfigurable Optical Interconnects — Jun. 2020

Invited Talk Optical/Photonic Interconnects for Computing Systems (OPTICS) Workshop — Dresden, Germany
Optimal Pairing and Non-Uniform Channel Alignment of Microring-Based Transceivers for Comb Laser–Driven DWDM Silicon Photonics — Mar. 2018

Seminar Department of ECE, Hong Kong University of Science and Technology — Hong Kong SAR, China
Variation-Aware Modeling and Design of Silicon Photonic Systems — Jan. 2018

Poster Optical/Photonic Interconnects for Computing Systems (OPTICS) Workshop — Lausanne, Switzerland
Variation-Aware Modeling and Design of Nanophotonic Interconnects — Mar. 2017